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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester(2025-26)
Faculty Name	Ms. Alluri Swathi
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Branch	CSE
Course Outcome	2

Case Study Topic: SecureVote – Digital Voting & Election Analytics System using Segment Tree

Work Done Summary:

Implemented a **Segment Tree** in Java for secure election analytics and vote monitoring. Stored vote counts of candidates in the Segment Tree for efficient range-based vote analysis. Inserted multiple voting records and updated vote counts dynamically during ballot casting. Implemented range query operations to analyze votes between candidate ranges and election regions. Supported fast vote updates, fraud detection, and auditing operations using Segment Tree properties. Segment Tree provides efficient performance for election result analysis and real-time vote monitoring systems.

Implementation Details :

1. Defined SegmentTree class with arrays: tree[], votes[] and integer n.
2. buildTree() constructs the Segment Tree from initial vote counts.
3. update() updates vote count dynamically when new votes are cast.
4. rangeQuery(left, right) retrieves total votes within a given candidate range.
5. Fraud detection performed by monitoring abnormal vote updates.
6. Main method inserted vote counts, updated votes, and executed range queries.
7. Segment Tree supports efficient analytics for large-scale election systems.

Result : Screenshots/Output

```
PS D:\JavaPrograms> javac SecureVoteSegmentTree.java
PS D:\JavaPrograms> java SecureVoteSegmentTree
Original Vote Counts:
Candidate 1 = 120
Candidate 2 = 150
Candidate 3 = 100
Candidate 4 = 180
Candidate 5 = 130

Segment Tree Array:
Tree[1] = 680
Tree[2] = 370
Tree[3] = 310
Tree[4] = 270
Tree[5] = 100
Tree[6] = 180
Tree[7] = 130
Tree[8] = 120
Tree[9] = 150

Total Votes from Candidate 2 to Candidate 4:
Range Query Result = 430

Updating Candidate 3 Votes to 170...

Updated Vote Counts:
Candidate 1 = 120
Candidate 2 = 150
Candidate 3 = 170
Candidate 4 = 180
Candidate 5 = 130

Segment Tree Array:
Tree[1] = 750
Tree[2] = 440
Tree[3] = 310
Tree[4] = 270
Tree[5] = 170
Tree[6] = 180
Tree[7] = 130
Tree[8] = 120
Tree[9] = 150

Range Query After Update:
Updated Range Sum = 500

No Fraud Detected.
```

Time Complexity:

- Build Segment Tree: **O(n)**
- Update Operation: **O(log n)**
- Range Query: **O(log n)**
- Segment Tree is highly efficient for dynamic vote updates and election analytics.

GitHub Link : https://github.com/tejasri3123/Dsa-2

Student Signature	**Faculty Signature**